

The Evolution of Machine Learning and AI

5 ECTS | Lecturer: Indrakshi Dey

- Department of Computing and Mathematics
- Intermediate level

Why this module matters

AI and ML are reshaping how software is built, how decisions are made, and how digital services are designed.

- From expert systems to LLMs
- From rule-based reasoning to learned representations
- From narrow tools to general-purpose intelligence

A brief historical arc

- 1950s-60s: symbolic AI and early logic-based systems
- 1970s-90s: expert systems, limited progress, and AI winters
- 2000s: data availability, statistical learning, and neural networks
- 2010s: deep learning, CNNs, RNNs, and transformer models
- 2020s: generative AI, LLMs, and responsible AI debates

Key themes in the module

- History of AI and machine learning
- Classical ML and early neural network foundations
- Training, optimisation, and deep learning breakthroughs
- Model families: CNNs, RNNs, transformers, diffusion models
- LLMs, software engineering workflows, and deployment
- Ethics, fairness, bias, explainability, and accountability

Learning outcomes

By the end of the module, students will be able to:

1. Trace the major milestones in AI and ML history.
2. Compare classical ML and modern deep learning approaches.
3. Evaluate modern LLM architectures and their significance.
4. Analyse the ethical and practical implications of AI systems.

Assessment and practice

- Continuous assessment: 100%
- Assignment 1: foundations, historical milestones, tools, and ethics
- Assignment 2: LLMs, workflows, and emerging AI trends
- Lectures, practical labs, and case-based learning

Looking ahead

The future of AI is not just about capability, but also trust, transparency, and responsible design.

This module introduces the ideas, tools, and critical thinking needed to navigate the rapidly evolving landscape of AI.